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Task: individual assignment 1

Q1. In a market research study analyzing customer preferences for a new beverage, data were collected on customer age, gender, preferred flavor (e.g., citrus, berry, or tropical), and the number of bottles purchased per month.

- a. Identify qualitative and quantitative variables from the above study. (2 marks)
 - i). qualitative variables
 - . customer gender
 - . customer preferred flavor
 - ii). quantitative variables
 - . customer age
 - . number of bottles purchased per month
- b. Define the population and sample from the study above. (2 marks)
 - (i). population
 - is the number of all customers who might buy the new beverage
 - (ii). Sample
 - Is the number of customers whose data were collected for age gender preferred flavor and number of bottles purchased per month

Q2. The data below shows the distribution of the number of hours students spent studying for an exam:

Number of hours	frequency
5 - 9	5
10 - 14	12
15 - 19	32
20 - 24	40
25 - 29	16
30 - 34	9
35 - 39	6

- a. How many students were surveyed in total? (1 mark)

Σf

$$5 + 12 + 32 + 40 + 16 + 9 + 6$$

$$= 120$$

- b. In which range of hours did most students study? (1 mark)

20 - 24

- c. If a student wants to be among the top 10% of the longest study times, how many hours should they aim to study at a minimum? Use the data to estimate your answer. (3 marks)

Solution

$$K = 10/100(120 + 1)$$

$$1/10(121)$$

$$12.1$$

Since 12.1 student fall within the range of 30 – 34 hours and above therefore the student should study at least 30 hours

- d. If a student wishes to understand the variability in the number of hours studied by their peers, how would they estimate how spread out the study hours are from the average? Hint: Compute the standard deviation. (6 marks)

class	f	x	fx	fx ²	Mean= ($\Sigma f x / \Sigma f$) ²
5 - 9	5	7	35	245	
10 - 14	12	12	144	1,728	
15 - 19	32	17	544	9,248	
20 - 24	40	22	880	19,360	
25 - 29	16	27	432	11,664	
30 - 34	9	32	288	9,216	
35 - 39	6	37	222	8,214	
	$\Sigma f=120$		$\Sigma fx=2,545$	$\Sigma fx^2=59,675$	$(2,545/120)^2 = 449.86$

$$S = \sqrt{1/n \cdot \Sigma fx^2 - (\text{mean})^2}$$

$$= \sqrt{1/120(59,675) - 449.86}$$

$$= \sqrt{497.29 - 449.86}$$

$$= \sqrt{47.43}$$

$$= 6.89$$

Q3. A survey was conducted on the number of books read by a group of students in a month, and the results are shown in the following stem-and-leaf plot:

Stem | Leaf

```

0 | 1 2 4 5 6 8
1 | 0 1 2 3 4 6 8
2 | 1 3 4 6
3 | 0 2 5 5

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Numbers: 1,2,4,5,6,8,10,11,12,13,14,16,18,21,23,24,26,30,32,35,35

a. What is the average number of books read by the students? (2 marks)

Average = total number of the books

Books (n)

$$= \frac{1 + 2 + 4 + 5 + 6 + 8 + 10 + 11 + 12 + 13 + 14 + 16 + 18 + 21 + 23 + 24 + 26 + 30 + 32 + 35 + 35}{21}$$

21

$$= \frac{346}{21}$$

21

$$= 16.48$$

b. Considering the data represented in the stem-and-leaf plot, what is the most frequently read number of books among the students? (2 marks)

from stem 3: | 0 2 5 5

$$= 3: 30 32 35 35$$

Therefore, the most frequently read number of the books is

$$= 35$$

- c. If a student wants to read more books than half of the students in the group, how many books should they aim to read in a month? (3 marks)

Solution

Sort the data set in ascending order

1,2,4,5,6,8,10,11,12,13,14,16,18,21,23,24,26,30,32,35,35
= 14 they should aim to read 14 books

- d. If a student wants to read as many books as the top 25% of students in the group, how many books should they aim to read in a month? (3 marks)

$K = 75/100 (21 + 1)$ th value

$= 3/4(22)$

$= 33/2$

$= 16.5^{\text{th}}$ value

Then we apply interpolation

$= 16^{\text{th}}$ value + $0.5(17^{\text{th}}$ value - 16^{th} value)

$= 24 + 0.5(26 - 24)$

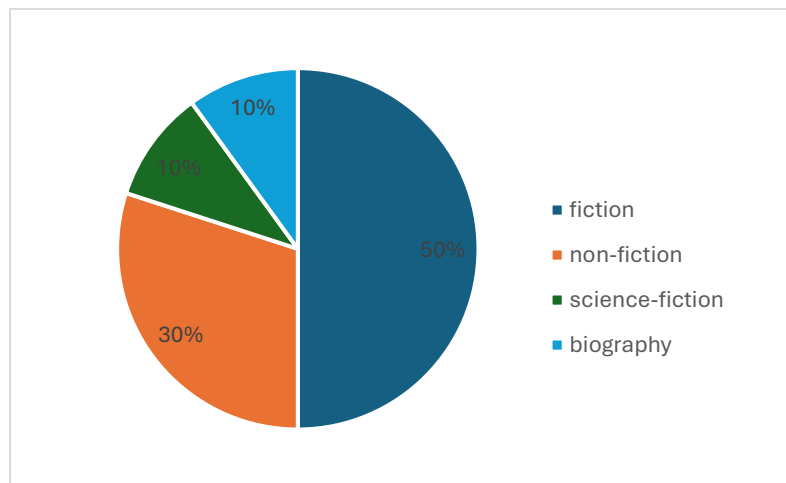
$= 24 + 0.5(2)$

$= 24 + 1$

$= 25$

Therefore, students should aim to read 25 books

Q4. The pie chart represents the types of books read by a group of book club members:



- a) If there are 40 members in the book club, how many members read non-fiction?
(1 mark)

Solution

non-fiction = 30%

40 members

$$= 30/100 \times 40$$

$$= 3/10 \times 40$$

$$= 3 \times 4$$

$$= 12$$

12 members read non-fiction

- b) If 5 new members join and prefer science fiction, how does this affect the percentage of science fiction readers? (2 marks)

Solution

science fiction = 10%

total number of members = 40

$$10\% \times 40$$

= 4 members read science – fiction

$$4 + 5 = 9$$

Now, the total number of members = 40 + 5

$$45$$

$$9/45 \times 100$$

$$= 20\%$$

therefore, the percentage of science-fiction readers will increase by 10%

- c. What is the proportion of members who read fiction compared to those who read nonfiction? (2 marks)

Solution

Fiction reader = 50%

$$50\% \times 40$$

$$= 20 \text{ members}$$

Non-fiction = 30%

$$30\% \times 40$$

$$12 \text{ members}$$

The proportion of fiction readers to non-fiction

$$= 20/12$$

$$5/3$$

$$= 5:3$$

